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DiabSecure: Federated-TinyML-enabled unified Privacy-Preserving Framework for Diabetes Risk Assessment using Zero-Trust Blockchain

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***Abstract*—In this paper, we introduce *DiabSecure*: a decentralized framework based on FL for real-time diabetes risk prediction using wearable healthcare devices. Local TinyML models deployed on each wearable device are trained on-device, with no need to share raw data, ensuring privacy and security. Model updates are aggregated using the popular FedAvg algorithm, achieving a global accuracy of 0.7593 and an F1-score of 0.7691. To further enhance performance, advanced aggregation strategies such as FedSGD and FedOpt are integrated, with FedSGD attaining an accuracy of 0.7563 and F1-score of 0.7692, and FOA achieving the highest accuracy of 0.76 and F1-score of 0.7725. Precision scores for FOA, FedAvg, and FedSGD are 0.774, 0.772, and 0.76 respectively, highlighting robust predictive power. Achieving this level of performance on the challenging dataset [1],which contains imbalanced, high-dimensional, and behaviorally diverse health records—is non-trivial. To ensure secure, tamper-proof updates, blockchain and IPFS manage data integrity and transparent logging. Our system operates entirely on-device using converted TFLite models, supports 6G-ready communication, and maintains continuous learning at the edge. *DiabSecure* stands out from traditional centralized systems by relying on dynamic, real-time patient data while ensuring privacy. Evaluation confirms the effectiveness of this approach in dynamic healthcare environments, with high precision and F1-scores across all algorithms, making it a promising solution for continuous health monitoring.**

***Index Terms*—Federated Learning, TinyML, Diabetes Risk Prediction, Zero-Trust Blockchain, InterPlanetary File System (IPFS), Decentralized Framework.**

I. Introduction

Currently, smart health devices and continuous patient monitoring have improved diabetes risk assessment. However, sending sensitive data over networks raises privacy concerns and exposes systems to misuse. Traditional, centralized models also suffer from scalability and securit issues [2]–[4]. Additionally, centralized models send patient data to servers, causing delays and raising security risks. Though machine learning improves predictions, it requires lots of data, leading to privacy concerns and high computational demands [5], [6]. To overcome these challenges, we introduce DiabSecure, a Federated-TinyML-enabled unified framework that ensures privacy-preserving diabetes risk assessment using a Zero-Trust Blockchain approach [7].

Recent studies in healthcare Industry 5.0 highlight the role of smart wearables and ML models like SVM, DT, and RF for activity prediction, but the use of TinyML for on-device, real-time health monitoring remains underexplored. This gap motivates further research on integrating TinyML with wearables for secure and efficient patient care. [8], [9]. FL helps secure healthcare data by training models on local devices without sharing raw data. Yet, its application in diabetes risk prediction on wearables is still new. DiabSecure addresses this by using FL to ensure accurate, secure diabetes risk assessment [10]–[12].

To enhance the security of DiabSecure even more, blockchain technology is incorporated to offer an extra layer of protection, transparency, and integrity [13], [14]. Taking advantage of the distributed and immutable nature of blockchain, all updates to the model and associated transactions are protected to avoid unauthorized changes. Every update is timestamped and kept on an immutable ledger so that no information can be adjusted or destroyed. Moreover, model updates and improvements are managed with an off-chain method such as IPFS , which improves security and system performance. By integrating blockchain with Federated Learning, DiabSecure preserves patient data privacy and ensures the diabetes risk assessment process integrity and trustworthiness.

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A. Novelty

The novelty of *DiabSecure* lies in its decentralized FL-based framework that enables secure, real-time diabetes risk prediction from wearable and IoT health devices. Unlike centralized approaches, *DiabSecure* trains TinyML models directly on-device and uses FedAvg, FedSGD, and FedOpt strategies to collaboratively update model weights. *DiabSecure* excels at real-time health data monitoring, a critical feature often overlooked by other models. Blockchain and IPFS integration ensure secure, tamper-proof weight sharing and protect patient data during training. With its low-latency edge deployment, *DiabSecure* guarantees efficient diabetes risk prediction without compromising privacy.

B. Research Contributions

Key contributions of this work include:

- DiabSecure Framework: A real-time, privacy-preserving diabetes risk prediction framework leveraging Federated-TinyML and blockchain.
- Lightweight TinyML with FL: On-device TinyML model training and collaborative global aggregation using FedAvg, FedSGD, and FedOpt to maintain data privacy and ensure secure, decentralized learning.
- Comprehensive Evaluation: Thorough testing of DiabSecure's accuracy, precision, and F1-score, showcasing superior performance and robustness compared to existing methods.

C. Article Layout

Rest of the paper is organized as follows. Section II explores the existing literature. Section III describes the system model. Section IV provides the problem formulation and loss functions. Section V discusses the privacy constraints. Section VI presents the proposed approach. Section VII contains the results and performance analysis of the proposed framework. Finally, Section VIII concludes the paper.

II. Literature Review

Recent studies have leveraged machine learning for diabetes risk prediction. Sun et al. 2025 [2] used 556,000 samples to evaluate six ML models, identifying glycated hemoglobin as a key predictor. Luz et al. 2025 [3] also showed that XGBoost performed well in diabetes risk prediction. Lei et al. 2025 [4] explored decision trees and ensemble techniques with PCA for better predictions. Xu et al. 2025 [5] integrated bioinformatics, statistics, and ML to highlight aging, obesity, and metabolic disorders as key risk factors. Khalifa et al. 2024 [6] emphasized that AI enhances diabetes care but requires robust security and patient-centered approaches.

Recent studies [7] use federated learning with blockchain, differential privacy, and homomorphic encryption to secure disease prediction from EHRs. However, they focus on large systems, lacking lightweight TinyML models for real-time edge healthcare. This research [15] presents a decentralized healthcare framework using federated learning and blockchain to protect EHR-based disease prediction. It ensures privacy and scalability with differential privacy, homomorphic encryption, and secure aggregation for safe, collaborative model training. Singh et al 2025 [12], The paper presents a federated learning-based DFU detection model combining segmentation and classification, using multimodal features from thermal and RGB images for precise early diagnosis. This system preserves data privacy and achieves high accuracy by leveraging transfer learning techniques.

Another work by Gudino et al 2024 [9],The study highlights the potential of CTGAN enhanced e-nose systems with TinyML integration for continuous monitoring of blood glucose levels (BGL).This study [13], [14] explores the transformative role of blockchain in diabetic patient care, focusing on data privacy, security, and interoperability. It emphasizes blockchain's potential to authenticate medical supply chains and provide emergency access to patient histories.

III. System Model

Fig. 1 shows the FL system for secure diabetes prediction. Health data from wearables is processed locally by TinyML models, which share only model weights with the global server. Blockchain and IPFS protect data, while FedAvg, FedSGD, and FedOpt update the global model. This setup ensures real-time, private, and reliable predictions at the edge.

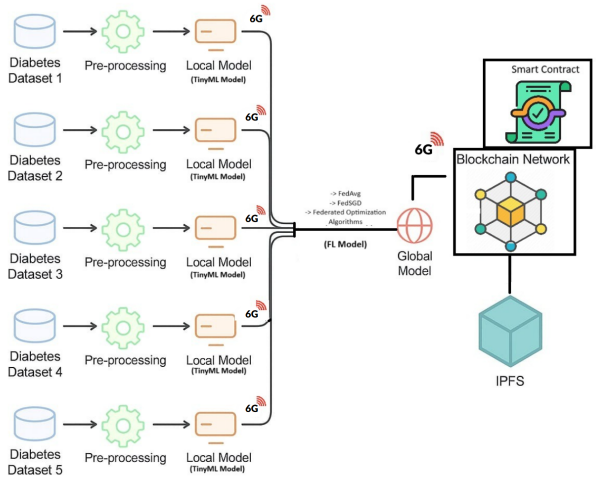

Fig. 1: DiabSecure architecture combining TinyML, federated learning, and blockchain for secure diabetes prediction.

The system comprises several key components:

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1) Data Collection and Pre-processing : Each wearable device gathers diabetes-related data, which is pre-processed to clean, normalize, and standardize it for machine learning models.
2) TinyML-enabled Local Models : After pre-processing, TinyML models are deployed on resource-constrained wearable devices. Each device independently trains a local model using its own dataset, enabling on-device diabetes risk prediction. This approach ensures low latency, real-time predictions, and enhanced data privacy since sensitive data never leaves the device.
3) Federated Learning : To improve model accuracy and maintain privacy, Federated Learning aggregates updates from local models using FedAvg and FedSGD, forming a secure, high-performing global model.
4) Blockchain and Smart Contracts : Blockchain ensures tamper-proof integrity of DiabSecure by recording updates and transactions on an immutable ledger, with smart contracts maintaining transparent and secure model management.
5) IPFS for Off-chain Storage : To reduce blockchain storage load, model updates and global model parameters are stored in IPFS off-chain, while the blockchain ledger records the IPFS hash pointers for secure and efficient data retrieval.
6) The process keeps going until the global model reaches the desired accuracy, making sure diabetes risk predictions stay secure and private.

IV. Problem Formulation and Loss Functions

The objective of all four algorithms is to collaboratively train models for diabetes prediction, ensuring privacy preservation and effective generalization. The primary focus is to minimize the loss function while achieving high predictive accuracy.

A. Algorithm 1: DIABSECURE – TinyML-Based Diabetes Prediction

Problem Formulation:

$$\min_{w} \frac{1}{N} \sum_{i=1}^{N} L(f(x_i; w), y_i) \tag{1}$$

where:

- N – number of samples in dataset D.
- x_i, y_i – input features and binary diabetes label for sample i.
- $f(x_i; w)$ – ANN-based TinyML model with weights w.

Loss Function: Binary Cross-Entropy

$$L(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

where:

- y – binary ground truth label.
- $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$ – sigmoid output.

Algorithm 1 DiabSecure: TinyML-Based Diabetes Prediction

procedure DiabSecure(D, E, B)
  Input: D - Dataset, E - Local training epochs, B - Batch size
  Output: Trained TinyML model M
  Load dataset D
  Separate features X and labels y
  Normalize features using *StandardScaler*
  Split (X, y) into training and test sets
  Initialize ANN model M with random weights
  for $e = 1$ to E **do** ▷ Training loop
    Train M on training data $(X_{\text{train}}, y_{\text{train}})$
    Compute loss:

$$L(y, \hat{y}) = -\frac{1}{n} \sum_{i=1}^{n} y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)$$

  end for
  Evaluate M on test set $(X_{\text{test}}, y_{\text{test}})$
  Predict label using Sigmoid:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

  Convert model M to TFLite format:

$$M_{\text{TFLite}} \leftarrow \texttt{TFLiteConverter}(M)$$

  Save M_{TFLite} to disk
  return M_{TFLite}
end procedure

B. Algorithm 2: Federated Training for Diabetes Prediction

Problem Formulation:

$$\min_{\theta} \frac{1}{K} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i=1}^{n_k} \mathcal{L}\big(f(x_i; \theta), y_i\big), \tag{2}$$

where:

- K denotes the number of participating clients in the federated learning framework.
- n_k represents the number of data samples on client k.
- (x_i, y_i) are the input features and the binary diabetes label for sample i in client k.
- $f(x_i; \theta)$ is the 2-layer neural network model parameterized by θ.

Loss Function: Binary Cross-Entropy

$$\mathcal{L}(y, \hat{y}) = -\big[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\big], \tag{3}$$

where:

- $y \in \{0, 1\}$ is the true label indicating diabetes presence.
- $\hat{y} = f(x; \theta)$ is the predicted probability of diabetes.

Federated Aggregation Strategies:

1) **FedAvg** In this method, local model parameters from each client are averaged to update the global model:

$$\theta^{t+1} = \frac{1}{K} \sum_{k=1}^{K} \theta_k^{t+1}.$$

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Algorithm 2 Unified Federated Learning for Diabetes Prediction (FedAvg, FedSGD, FedOpt)

Input: Preprocessed dataset D, number of clients K, local epochs E, communication rounds T, FL variant $\in$ {FedAvg, FedSGD, FedOpt}
Output: Final global model $\mathcal{M}(\theta_T)$ and evaluation metrics
Normalize features: $X \leftarrow \frac{X-\mu}{\sigma}$
Split D into K disjoint client datasets: $D_1, D_2, \ldots, D_K$
Define model $\mathcal{M}$:
    $h = \text{ReLU}(W_1 x + b_1)$
    $\hat{y} = \sigma(W_2 h + b_2)$ ▷ Sigmoid for binary classification
Initialize global parameters θ_0 from $\mathcal{M}$
for $t = 1$ to T **do**
    Broadcast θ_{t-1} to all clients
    for each client $k = 1$ to K **in parallel do**
        Set local model $\theta_k^t \leftarrow \theta_{t-1}$
        if FL variant == FedSGD **then**
            Sample one minibatch $(x_i, y_i) \in D_k$
            Forward: $\hat{y}_i = \mathcal{M}(x_i)$
            Compute loss: $\mathcal{L}_i = -[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$
            Backpropagate and update: $\theta_k^t \leftarrow \theta_k^t - \eta \nabla_\theta \mathcal{L}_i$
        else if FL variant == FedAvg or FedOpt **then**
            for $e = 1$ to E **do**
                for each minibatch $(x_i, y_i) \in D_k$ **do**
                $\hat{y}_i = \mathcal{M}(x_i)$
                Compute loss: $\mathcal{L}_i = -[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$
                Backpropagate and update: $\theta_k^t \leftarrow \theta_k^t - \eta \nabla_\theta \mathcal{L}_i$
                end for
            end for
        end if
        Evaluate local model on D_k: compute Accuracy$_k$ and F1$_k$
        Return updated θ_k^t, Accuracy$_k$, and F1$_k$ to server
    end for
    if FL variant == FedAvg or FedSGD **then**
        Aggregate: $\theta_t \leftarrow \frac{1}{K}\sum_{k=1}^{K} \theta_k^t$
    else if FL variant == FedOpt **then**
        Use optimizer (e.g., Adam) to update $\theta_t \leftarrow \text{Opt}(\theta_{t-1}, \{\theta_k^t\})$
    end if
    Evaluate global model $\mathcal{M}(\theta_t)$ on hold-out set D_0
    Compute and store Accuracy, F1-score
end for
return $\mathcal{M}(\theta_T)$ and evaluation metrics

2) **FedSGD** Here, gradients computed locally at each client are averaged and used to update the global model:

$$\theta^{t+1} = \theta^t - \eta \frac{1}{K} \sum_{k=1}^{K} \nabla \mathcal{L}_k(\theta^t),$$

where η denotes the learning rate and $\nabla \mathcal{L}_k(\theta^t)$ is the gradient of the local loss for client k.

3) **FedOpt** A generalized framework for server-side aggregation using advanced optimizers such as Adam, Yogi, or momentum-based methods:

$$\theta^{t+1} = \text{Optimizer}\big(\theta^t, \{\theta_k^{t+1}\}_{k=1}^{K}\big),$$

where Optimizer$(\cdot)$ performs the parameter update based on the aggregated client updates.

This formulation enables collaborative learning for diabetes prediction while preserving client data privacy.

V. Privacy Constraints

The *DiabSecure* framework makes sure that health wearables do not exchange raw health information, thus ensuring patient privacy and data integrity. Rather, every device trains a TinyML-based local model from its own data and only sends encrypted model updates (weights) to the federated learning server. The following practices are used to ensure privacy protection and adversarial attacks against model aggregation :

- **Secure Aggregation:** The FL server aggregates the model updates from various wearable devices using *FedAvg* and *FedSGD*, ensuring that individual client data remains private and is never exposed during the aggregation process.
- **Blockchain-based Integrity Verification:** Model updates are applied to the blockchain in an IPFS hash and can be kept tamper-proof during the weight exchanges while making the unauthorized modifications difficult and ineffective.

VI. Proposed Approach

In this section, we outline the heart and soul of the DiabSecure framework—a privacy-respecting pipeline that blends TinyML, FL, and Zero-Trust principles to assess diabetes risk without compromising user data.

A. Data Preprocessing and Local TinyML Model Training

The journey begins at the client side. Using the publicly available [1] dataset, we performed standard preprocessing steps including feature scaling via StandardScaler and a stratified split into training and testing sets.

Each client is equipped with a compact, resource-efficient deep learning model—essentially a TinyML model. Specifically, the local model comprises a simple architecture with an input layer, one dense hidden layer with ReLU activation, and a final sigmoid output layer for binary classification (Diabetic / Non-Diabetic). Despite its simplicity, this model achieved competitive results after training using the Adam optimizer and binary cross-entropy loss.

To ensure robustness, we employed a custom callback during training to monitor the F1-score across epochs, which serves as a more balanced evaluation metric than accuracy in this healthcare context. Post training, each TinyML model is converted to the TFLite format, ensuring compatibility with edge devices like smartphones or microcontrollers.

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B. Federated Learning Integration

TinyML models are powerful, but their true strength shines when they collaborate. We introduced Federated Learning to enable decentralized training across multiple clients while keeping sensitive data local. The simulation involved three FL clients and one central server.

We evaluated three Federated Learning algorithms:

- **FedAvg** – computes a weighted average of model weights from each client to form the global model.
- **FedSGD** – a variation of FedAvg using stochastic gradient descent on client-side updates.
- **Federated Optimization Algorithm** – a custom optimizer-based approach that builds on client weight contributions using local gradients.

Each client trains its local TinyML model for a fixed number of epochs. After training, only the model weights are transmitted to the central server—no raw data ever leaves the client device. The server then performs aggregation according to the selected FL strategy. This process is repeated for multiple rounds (typically 3 in our setup), gradually refining the global model.

The entire FL workflow for each round proceeds as follows:

1) Each client trains its TinyML model locally.
2) Model weights are sent to the central server.
3) The server aggregates weights using FedAvg, FedSGD, or the optimizer-based method.
4) The updated global model is redistributed to all clients.

The training logs showed that FedAvg offered the most stable performance across rounds, while FedSGD was slightly faster but more variable in outcome.

C. Blockchain and Privacy

To reinforce privacy and prevent model poisoning or malicious injections, we embedded a Zero-Trust Blockchain layer within the federated setup. Each transaction—be it local model updates or global model dissemination—is recorded immutably, ensuring verifiability and transparency across the system.

Moreover, each node is verified before participation, and model updates are accepted only after cryptographic validation, maintaining the zero-trust ideology.

D. System Flow Summary

The entire DiabSecure framework follows this pipeline:

1) Local devices train TinyML models on personal data.
2) Model weights (not data) are shared securely with the federated server.
3) Global model is computed via aggregation (FedAvg, FedSGD, or Federated Optimization Algorithm).
4) Blockchain ensures transactional integrity across model communications.
5) The improved model is sent back to the devices, forming a feedback loop.

E. Why This Matters

This human-centric architecture empowers diabetic risk prediction at scale while preserving individual privacy. Instead of centralizing sensitive health data, we put intelligence directly into the user's hands (or devices) and build a smarter, unified system—together, yet apart.

VII. Experimental Results

The results evaluation section assesses the performance of the proposed *DiabSecure* framework by comparing its accuracy, loss, and F1-score across different federated learning strategies, including FedAvg, FedSGD, and Federated Optimization Algorithm. Additionally, the framework's decentralized TinyML-based approach is benchmarked against traditional centralized models to highlight its effectiveness, scalability, and privacy-preserving capabilities in diabetes risk prediction.

A. Experimental Setup and Dataset Description

1) Experimental Setup: The implementation of the DiabSecure framework was conducted on a Windows 11 machine configured with an Intel Core i7 processor, 16GB of RAM, and an NVIDIA GeForce RTX 4050 GPU with 8GB of VRAM.

2) Dataset Description: The dataset [1] used in this research focuses on diabetes risk assessment and comprises a mix of medical history, lifestyle choices, and demographic information. Key features include blood pressure status (*HighBP*), cholesterol levels (*HighChol*), BMI, smoking habits, physical activity, general health perception, and history of cardiovascular conditions such as stroke and heart disease.

In addition to medical indicators, the dataset also incorporates social and economic factors such as education level, income group, access to healthcare, and whether the individual had to skip medical visits due to cost concerns. The target variable, *Diabetes_012*, reflects the individual's diabetes status or risk level.

Altogether, the dataset [1] provides a well-rounded view of each individual's health and lifestyle, making it suitable for training privacy-preserving federated models for diabetes prediction.

B. Performance Metrics

To evaluate the performance of the DiabSecure framework, several standard classification metrics were considered:

- **Confusion Matrix:** A table summarizing the number of correct and incorrect predictions made by the model, categorized by actual and predicted classes. It provides a clear view of true positives, false positives, true negatives, and false negatives.
- **Accuracy:** Represents the overall proportion of correctly classified instances among all predictions. It gives a general sense of the model's performance across all classes.
- **Loss:** The loss function, typically binary cross-entropy in this context, measures how far the predicted probabilities

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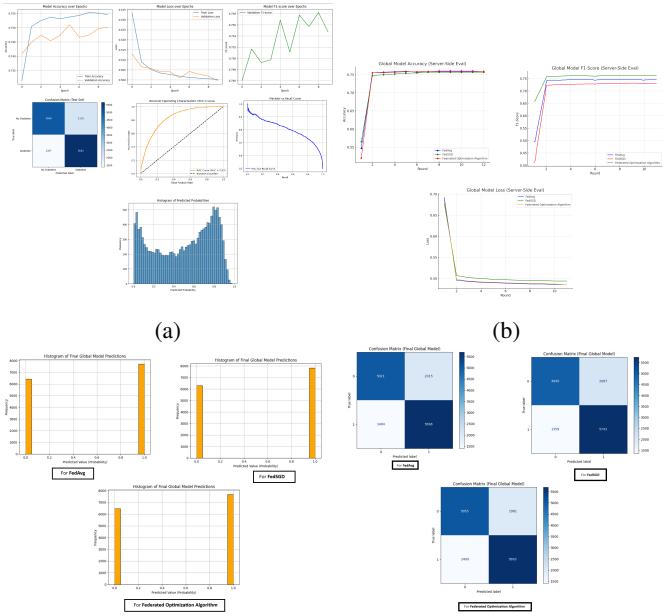

(a) (b) (c) (d)

Fig. 2: (a) Result Graphs for TinyML Model , (b) Accuracy, F1 Score and Loss curves of all FL Models used, (c) Histogram of Final Model Prediction for all FL Models used, and (d) Confusion Matrix of all FL Models used

are from the actual labels. Lower loss values indicate better model performance.

- **F1-Score:** The harmonic mean of precision and recall, used to balance false positives and false negatives, especially when class distribution is imbalanced. It provides a more comprehensive measure than accuracy alone in binary classification tasks.

These metrics ensure a holistic evaluation of the framework's effectiveness in predicting diabetes risk while maintaining data privacy in a federated learning setup.

C. Result and Analysis

This section evaluates the performance of the DiabSecure framework under various federated learning strategies and compares it with a traditional TinyML model. The analysis is based on metrics such as accuracy, loss, F1-score, and confusion matrices observed during server-side evaluations. Figures 2a, 2b, 2c, and 2d illustrate the trends and outcomes of each model configuration.

1) TinyML Model: As illustrated in the system architecture Fig. 1, each edge device or local healthcare unit trains a TinyML model independently on its respective diabetes dataset. This mirrors real-world scenarios where data remains decentralized across hospitals, clinics, or IoT-enabled health monitors. The use of 6G connectivity enables timely and efficient communication without compromising latency or device limitations.

As seen in Fig. 2a, the TinyML model shows a promising trend in accuracy, reaching a stable level around 0.755. Both

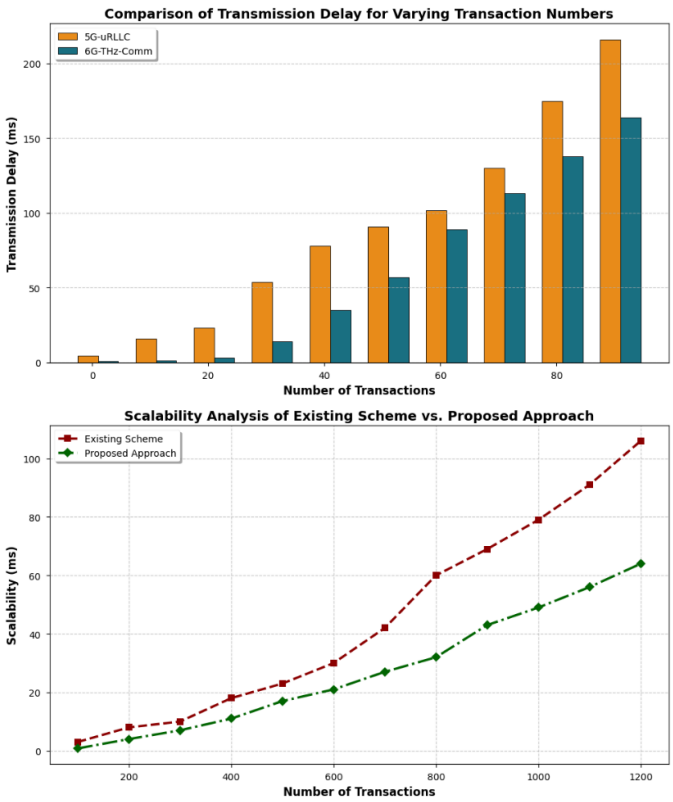

Fig. 3: (a) Transmission Delay Performance for Blockchain Transactions Across 5G-uRLLC and 6G-THz-Comm and (b) Scalability Performance Comparison of Proposed Blockchain Approach Against Existing Scheme

training and validation losses consistently decrease across epochs, which reflects strong generalization capability.

The F1-score also trends upward, indicating that the model balances precision and recall well. The confusion matrix confirms the model's effectiveness in classifying diabetic and non-diabetic patients, while the ROC curve (AUC = 0.83) emphasizes its strength in distinguishing between classes. The precision-recall curve and probability histogram further demonstrate balanced performance.

The overall performance is summarized in Table I, where we see accuracy improving round-by-round and loss gradually reducing, with the F1-score stabilizing around 0.75, reflecting the model's consistency.

Overall, TinyML serves as a foundational stage, where local models contribute knowledge while preserving privacy. These models later participate in federated aggregation, as part of the DiabSecure pipeline.

2) FL using FedAvg Algorithm: After the TinyML models complete their local training, the FedAvg algorithm aggregates them by averaging their weights—enabling collaborative learning without sharing raw data. This makes FedAvg the privacy-preserving engine behind DiabSecure's federated architecture.

As depicted in Fig. 2b, the model achieves strong performance early, with accuracy rising above 0.75 within the

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first few rounds and maintaining that level consistently. The F1-score follows a similar trend, highlighting balanced and reliable predictions, while the loss curve drops sharply before stabilizing—indicating efficient convergence.

Prediction confidence is illustrated in Fig. 2c, where outputs are clustered near 0 and 1, demonstrating the model's certainty. Fig. 2d further confirms solid classification performance with a confusion matrix showing high true positive and true negative counts and few misclassifications.

Table II reinforces these insights: the accuracy peaks at 0.7593, loss continues to decline across rounds, and F1-scores remain in the 0.76–0.77 range—underscoring FedAvg's ability to deliver stable, high-quality results.

Overall, FedAvg proves to be a dependable and privacy-preserving choice for DiabSecure, balancing performance and simplicity.

3) FL using FedSGD Algorithm: FedSGD follows the same privacy-preserving framework as FedAvg but updates the global model using averaged gradients instead of weights. This allows for more frequent updates, which can be beneficial for responsiveness but introduces a bit more noise into the learning process.

As illustrated in Fig. 2b, FedSGD leads to a rapid rise in accuracy early on, eventually stabilizing around 0.755. The F1-score also shows strong, consistent performance with only slight variability—reflecting a solid trade-off between precision and recall. The global loss decreases steadily across rounds, indicating effective optimization despite the gradient-level updates.

In Fig. 2c, the prediction histogram reveals confident outputs clustered at the extremes (0 and 1), though slightly more scattered than FedAvg—suggesting marginally reduced certainty. The confusion matrix in Fig. 2d reinforces this, showing reliable class separation with a few additional false positives and negatives compared to FedAvg.

Table III quantifies this trend well: accuracy grows from 0.5615 to 0.7563, while F1-score steadily improves, eventually peaking around 0.77.

Overall, FedSGD offers a compelling option for DiabSecure when real-time adaptability is critical, delivering strong performance with quicker, more dynamic updates.

4) FL using Federated Optimization Algorithm: The FOA represents the most advanced aggregation strategy in the DiabSecure pipeline, using adaptive techniques and dynamic weighting to guide learning more intelligently.

As shown in Fig. 2b, FOA delivers highly stable performance—accuracy stays consistently above 0.76 with minimal fluctuation, and the F1-score surpasses all other approaches. The global loss quickly converges around 0.49, indicating efficient learning dynamics.

In Fig. 2c, the histogram of predictions reflects strong model confidence, with outputs sharply clustered at 0 and 1. The confusion matrix in Fig. 2d further supports this, showing excellent class separation and minimal misclassifications.

According to Table IV, FOA reaches an accuracy of 0.7602 by the final round, with steady improvements in both loss and F1-score throughout.

Overall, FOA stands out by combining precision, consistency, and confidence—making it the most robust federated learning strategy within the DiabSecure framework.

TABLE I: Results of TinyML Model Performance across Rounds

Round	Accuracy	Loss	F1 Score
1	0.6965	0.5707	0.7460
2	0.7525	0.5058	0.7517
3	0.7510	0.5077	0.7493
4	0.7541	0.5021	0.7497
5	0.7516	0.5019	0.7568
6	0.7574	0.4980	0.7511
7	0.7508	0.5062	0.7577
8	0.7572	0.4984	0.7557
9	0.7532	0.5012	0.7582
10	0.7540	0.5018	0.7547
TOTAL	0.7532	0.5015	0.75

TABLE II: Results of Model Performance using FedAvg Algorithm across Rounds

Round	Accuracy	Loss	F1 Score
0	0.5467	0.6924	0.4590
1	0.7549	0.4976	0.7672
2	0.7561	0.4952	0.7671
3	0.7568	0.4935	0.7692
4	0.7589	0.4927	0.7710
5	0.7570	0.4923	0.7681
6	0.7587	0.4913	0.7698
7	0.7589	0.4910	0.7706
8	0.7592	0.4906	0.7719
9	0.7593	0.4904	0.7683
10	0.7582	0.4899	0.7692

TABLE III: Results of Model Performance using FedSGD across Rounds

Round	Accuracy	Loss	F1 Score
0	0.5615	0.6782	0.3540
1	0.7484	0.5054	0.7595
2	0.7506	0.5017	0.7631
3	0.7521	0.4996	0.7641
4	0.7539	0.4981	0.7661
5	0.7547	0.4970	0.76631
6	0.7564	0.4961	0.7682
7	0.7559	0.4958	0.7692
8	0.7565	0.4952	0.7701
9	0.7562	0.4948	0.7696
10	0.7563	0.4944	0.7692

This strategy, combined with blockchain-backed integrity via smart contracts and off-chain IPFS storage (as shown in Fig. 1), ensures that the DiabSecure system remains privacy-preserving, transparent, and robust.

5) Blockchain Integration in DiabSecure: In the DiabSecure framework, blockchain technology plays a pivotal role in ensuring secure, tamper-proof, and decentralized management of FL updates. Our architecture leverages a **Zero-**

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TABLE IV: Results of FL Model Performance using Federated Optimization Algorithm across Rounds

Round	Accuracy	Loss	F1 Score
0	0.5189	0.6882	0.6398
1	0.7551	0.4974	0.7661
2	0.7563	0.4945	0.7679
3	0.7570	0.4933	0.7702
4	0.7578	0.4922	0.7703
5	0.7585	0.4920	0.7682
6	0.7595	0.4908	0.7715
7	0.7591	0.4903	0.7710
8	0.7587	0.4902	0.7711
9	0.7597	0.4892	0.7702
10	0.7602	0.4887	0.7706

Trust Blockchain Network integrated with **IPFS** to store and manage FL model parameters securely.

Figure 1 illustrates the interaction between distributed TinyML-based local models and the global model via high-speed 6G communication. Each participant trains a local model on a private diabetes dataset, and updates are communicated to a global model aggregator through federated learning algorithms such as FedAvg, FedSGD, and Federated Optimization Algorithm.

These updates are then verified and stored using a smart contract on a blockchain ledger, ensuring immutability and auditability of every model transaction. IPFS is employed for distributed off-chain storage, optimizing space and access efficiency while maintaining the integrity of shared FL models.

Advantages of Blockchain Integration:

- **Security and Trust**: By using a zero-trust blockchain framework, DiabSecure eliminates the need for centralized trust authorities, ensuring that only authenticated model updates are accepted.
- **Transparency**: Each local update is verifiable and auditable, enhancing trust among distributed stakeholders.
- **Data Privacy**: No raw health data is ever shared; only encrypted model weights are recorded on-chain, aligning with federated learning's privacy-preserving principles.
- **Resilience and Traceability**: Model updates are immutable and traceable, allowing for post-hoc accountability in healthcare applications.

This combination of blockchain and FL, supported by 6G connectivity, enables real-time and secure collaboration among decentralized health data contributors while adhering to privacy regulations.

VIII. Conclusion

In this paper, we introduced *DiabSecure*, a decentralized framework that integrates TinyML, FL, and blockchain-based IPFS to deliver secure and privacy-preserving diabetes risk prediction. TinyML models trained on wearable devices achieved an accuracy of 0.7574, while FL with FedAvg reached 0.7593, FedSGD 0.7563, and the federated optimization framework attained 0.76. Our approach ensures real-time, low-latency predictions without sharing sensitive patient data. The blockchain and IPFS ensure data integrity and secure model updates. Overall, DiabSecure demonstrates an effective and trustworthy solution for diabetes risk assessment in modern healthcare systems.

In the future, we plan to explore advanced FL algorithms such as FedProx and FedMA to further improve the model's accuracy and scalability. Incorporating personalized FL and meta-learning will enable dynamic adaptation for each patient's unique health patterns. We also aim to enhance the TinyML model's efficiency for real-time predictions on resource-constrained devices. These steps will ensure DiabSecure evolves into a more robust and adaptive diabetes risk prediction system.

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